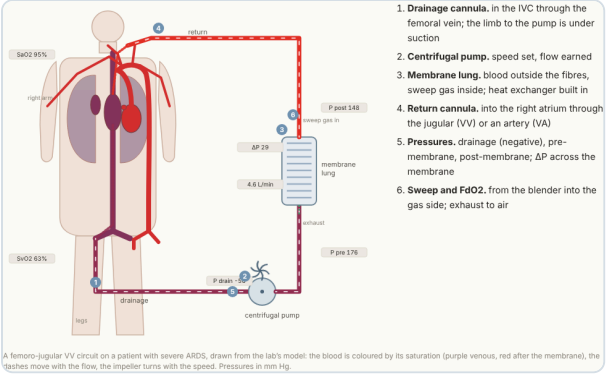
# ECMO bedside reference

VV and VA · circuit numbers · physiology · selection · anticoagulation · emergencies · liberation. After the ELSO guidelines (VV 2021, VA 2021, anticoagulation 2021/22, ECPR 2021, neurology 2024, hypothermia 2025) and the trials; companion to the PFT Interpreter's ECMO Learn area (v7.11). Educational use — the ECMO team and the local protocol run the circuit.




1. **Drainage cannula.** in the IVC through the femoral vein; the limb to the pump is under suction
2. **Centrifugal pump.** speed set, flow earned
3. **Membrane lung.** blood outside the fibres, sweep gas inside; heat exchanger built in
4. **Return cannula.** into the right atrium through the jugular (VV) or an artery (VA)
5. **Pressures.** drainage (negative), pre-membrane, post-membrane; ΔP across the membrane
6. **Sweep and FdO2.** from the blender into the gas side; exhaust to air

A femoro-jugular VV circuit on a patient with severe ARDS, drawn from the lab's model: the blood is coloured by its saturation (purple venous, red after the membrane), the dashes move with the flow, the impeller turns with the speed. Pressures in mm Hg.

## THE PRESSURES

| Pressure | Where | Typical | What it means when it moves |
|---|---|---|---|
| **Drainage (P1, venous, inlet)** | Between the drainage cannula and the pump | −20 to −80 mm Hg; keep above −100 | More negative: the pump is asking more than the cava gives — hypovolaemia, a kinked or migrated cannula, cough, tamponade, pneumothorax, abdominal pressure, too much speed. Past −100: haemolysis and vessel collapse (suck-down). |
| **Pre-membrane (P2, internal)** | Between the pump and the oxygenator | 100–250 mm Hg; below 300 | Rises with flow and with the resistance downstream: a clotting membrane or a narrow return. |
| **Post-membrane (P3, return, arterial)** | After the oxygenator | < 300 mm Hg | High: the return cannula (too small, kinked, clotted) or, on VA, the arterial pressure. |
| **ΔP (transmembrane)** | P2 − P3 | 10–40 mm Hg at 4–5 L/min; it rises with flow, so compare at the same flow | A rising ΔP at the same flow is clot in the fibres; > 60 and rising, or > 100, is a membrane that needs changing — with the gas transfer and the haemolysis markers, not alone. |

## IS THE MEMBRANE WORKING?

- Post-membrane gas on FdO2 1.0: PO2 > 250–300 on a new membrane; < 200 is failing (< 150 is failed). PCO2 should be well below the pre-membrane value (a difference < 10 mm Hg with adequate sweep = CO2 transfer failure).
- Oxygen transfer: 13.4 × flow (L/min) × Hb × (post − pre saturation) in mL/min; < 100–150 mL/min at full flow is a dying membrane.
- ΔP at a known flow, trended; visible clot on the inlet face; rising plasma free Hb, LDH, D-dimer, falling platelets and fibrinogen (consumption).
- Plasma leak (yellow fluid from the gas exhaust) and gas-side water: the membrane is wet; a brief increase of sweep clears condensation (a "sigh").
- A sudden desaturation with a normal ΔP: look at the gas line and the blender before the membrane.

## THE TWO DIALS

- Blood flow is the oxygen dial. Oxygen transfer = flow × (post − pre content) × 10; with a working membrane every millilitre that passes leaves fully saturated, so the patient's oxygenation is decided by how much of the venous return passes (VV) or by where the blood goes (VA). Targets: 60–80 mL/kg/min, or 3 L/min/m² (VV: about 60% of the cardiac output or more; VA: enough for the perfusion — CI > 2.2, SvO2 > 65%, lactate falling).
- Sweep gas is the CO2 dial. CO2 crosses the membrane some 20 times more easily than oxygen, so the clearance depends on the gas that carries it away: start at about half the blood flow (1:1 for VA aiming at normocapnia) and titrate to the PaCO2. New sweep ≈ current sweep × (current PaCO2 / target PaCO2). A sweep change takes 30–50 min to show fully in the arterial gas.
- FdO2 is for the post-membrane PO2, usually 1.0 to start; on VA (and after arrest) wean it so the arterial PaO2 stays below 300 — hyperoxia harms the reperfused brain and heart.
- Correct hypercapnia slowly: no more than about half of the excess in the first few hours, longer after chronic retention — a rapid fall in PaCO2 on cannulation is the setting of intracranial haemorrhage (ELSO neurological guidance: avoid rapid changes in the first 24 h).

## WHAT SETS THE SATURATION ON VV

- CaO2 = 1.34 × Hb × SaO2 + 0.003 × PaO2 (mL/dL); DO2 = CO × CaO2 × 10 (mL/min); VO2 = CO × (CaO2 − CvO2) × 10. Target DO2:VO2 ≥ 3–5; below 2 the tissues are supply-dependent and the lactate rises.
- Membrane oxygen transfer = Q × (Cpost − Cpre) × 10 ≈ 13.4 × Q × Hb × (Spost − Spre). It is largest when the drainage blood is desaturated: a pre-membrane saturation of 85% means either recirculation or a patient who is not extracting.
- Effective flow = pump flow × (1 − recirculation fraction). Recirculation R ≈ (Spre − Sv) / (Spost − Sv), where Sv is the true mixed venous saturation (e.g. from a PA catheter) — in practice: a drainage saturation above 75–80% with a falling SpO2, bright blood in the drainage line, and a paradoxical rise in SpO2 when the flow is turned down.
- Arterial saturation (no native lung contribution) ≈ [Qeff × Spost + (CO − Qeff) × Sv] / CO.

## HYPOXAEMIA ON VV — THE ORDER

1. Is it real? Co-oximetry on an arterial sample; perfusion, lactate, Hb.
2. The gas: oxygen supply, blender, sweep line attached, FdO2 1.0. (Thirty seconds; the commonest reversible cause.)
3. The membrane: post-membrane PO2 at FdO2 1.0 (> 200), ΔP trend at this flow, visible clot, consumption and haemolysis labs.
4. The flow: actual flow versus cardiac output (echo), drainage pressure, chatter; can it go up without suck-down?
5. Recirculation: drainage saturation, cannula positions on X-ray/echo, response to a small flow reduction.
6. The patient: fever, shivering, agitation, pain, seizures (VO2); anaemia; sepsis and a high output; new lung pathology (pneumothorax, effusion, oedema, mucus plug); the native lung's FiO2 and PEEP.
7. Then act in order of cost: treat demand, transfuse to Hb 9–10, deepen sedation ± paralysis, cool, optimise the native lung modestly (FiO2 up to 0.6, PEEP, inhaled vasodilator), more flow if drainage allows, a second drainage cannula (VV-V), reposition for recirculation; a β-blocker only as an expert, delivery-measured last resort. Accept SaO2 85–88% with normal lactate and organs; it is delivery that counts.

## LUNGS AT REST (ELSO 2021)

| Setting | Recommended | Acceptable | Why |
|---|---|---|---|
| **Plateau** | < 25 cm H2O (lower if possible) | ≤ 30 | The reason for ECMO in ARDS is to stop the ventilator injuring the lung |
| **PEEP** | ≥ 10 cm H2O | 10–24 | Keep the lung open without stretching it; higher if it recruits |
| **Rate** | 4–15/min | 4–30 | Less strain per minute; the sweep does the CO2 |
| **FiO2** | As low as the saturation allows | 0.3–0.5 | Oxygen toxicity and absorption atelectasis |
| **Tidal volume** | Whatever the pressures allow — often 2–4 mL/kg | ≤ 6 mL/kg | Driving pressure ≤ 15 as a pragmatic ceiling |
| **Effort** | Light sedation if the drive is gentle; deep sedation ± paralysis if the patient fights or pulls big breaths | | Patient self-inflicted lung injury: a high drive despite a good gas is still a strain |

## VV TARGETS

| Variable | Target |
|---|---|
| **SaO2 / SpO2** | ≥ 85–88% (PaO2 55–80) with normal lactate and organs; check SpO2 against co-oximetry |
| **Pre-membrane (drainage) saturation** | 60–75%; > 80% = recirculation or low extraction |
| **Post-membrane PO2 on FdO2 1.0** | > 250–300 new; < 200 failing |
| **PaCO2 / pH** | Patient's baseline; pH 7.35–7.45; slow correction |
| **Hb** | ≥ 7–9 g/dL (the Red Book's 12–14 is history; higher if delivery is short) |
| **Flow** | 60–80 mL/kg/min; ≥ 60% of the cardiac output; never below 2–2.5 L/min |
| **Fluids** | Negative balance once resuscitated; a dry lung recovers |

## VA HAEMODYNAMICS

- Total systemic flow = circuit flow + native ejection. MAP = total flow × SVR: the pump makes a pressure, and a vasoplegic patient still needs a vasopressor; a vasoconstricted one a vasodilator. Target MAP 65–80 (≥ 60 during weaning), the lowest that perfuses — every mm Hg above is afterload.
- Flow target: cardiac index > 2.2 L/min/m², SvO2 > 65%, lactate clearing, urine; usually 3–4.5 L/min; "partial support" (60–80% of the output) leaves the heart some work and keeps the valve opening.
- Pulse pressure is the ventricle's voice. ≥ 10–15 mm Hg means the valve opens and the LV ejects; < 10 (< 15 ≈ native output < 1 L/min) means a silent ventricle — distended, or empty (RV failure, hypovolaemia, tamponade): the echo decides which.
- The right heart is unloaded (the drainage takes its venous return); the left heart is loaded. A dilated RV on VA is tamponade, PE or a cannula problem until proven otherwise.
- Pulmonary blood flow is only what the RV still pumps: the native lung sees a fraction of the output, so end-tidal CO2 falls with native flow (ETCO2 < 15 ≈ native output < 1 L/min).

## LV DISTENSION

| | |
|---|---|
| **Why** | The LV must generate a pressure above the aortic pressure to eject; with a weak LV and a high circuit flow the valve stays shut, while bronchial and Thebesian flow, residual RV output and any aortic regurgitation keep filling the left heart. LVEDP and LA pressure climb; wall stress and subendocardial ischaemia prevent recovery; the lung floods (oedema, haemorrhage); stasis makes thrombus in the LV and the aortic root. |
| **Signs** | Pulse pressure < 10, aortic valve opening less than every 3 beats or not at all, LV dilating, spontaneous echo contrast, functional MR, rising PAOP/PA diastolic, pulmonary oedema or pink froth, ventricular arrhythmia. |
| **First measures** | Reduce the flow to the lowest that perfuses; afterload reduction; an inotrope (dobutamine, milrinone) to open the valve; PEEP for the oedema; avoid volume; full anticoagulation for the stasis. |
| **Venting** | Impella (transaortic microaxial pump; "ECPELLA") drains the LV directly — not with LV thrombus or a mechanical aortic valve; IABP lowers afterload and adds pulsatility (indirect, modest); percutaneous atrial septostomy / LA cannula; surgical LV apical or pulmonary venous vent; central cannulation with a vent. No single strategy is proven prophylactic; most centres vent when signs appear or early in a non-ejecting ventricle. |

## DIFFERENTIAL HYPOXAEMIA (HARLEQUIN)

| | |
|---|---|
| **Three ingredients** | Peripheral (femoral) arterial return; a native heart good enough to eject toward the arch; lungs bad enough that what it ejects is desaturated. Typical at day 2–5 as the myocarditis or infarct recovers while the lungs are wet. |
| **Who suffers** | The first branches from the root: the coronaries, the innominate (right arm, right brain), then the left carotid. The legs are pink. A femoral gas is a lie about the brain. |
| **Detection** | Right radial arterial line and a right-hand pulse oximeter from the first hour; cerebral NIRS; compare right radial with the post-membrane gas. The right radial PaCO2 is the native lung's, the post-membrane PCO2 is the sweep's: adjust sweep to the territory that matters and do not chase one sample. |
| **Treatment** | Treat the lungs (PEEP, FiO2, diuresis, secretions; distension causing the oedema?); more circuit flow pushes the zone proximally but adds afterload; add a venous return limb (V-AV: a jugular return cannula on a Y) so oxygenated blood enters the right heart and the native ejection is red; or move the arterial return to the axillary artery or central; if the heart is recovered, convert to VV. |

## VA TARGETS

| Variable | Target |
|---|---|
| **MAP** | 65–80 mm Hg (≥ 60 when weaning; > 70 for neuroprotection) |
| **Flow** | CI > 2.2; SvO2 > 65%; usually 3–4.5 L/min; partial support preferred |
| **Pulse pressure** | ≥ 10–15 mm Hg; aortic valve opening |
| **Right radial PaO2** | 70–150 (< 300); SaO2 92–97% after arrest |
| **Post-membrane PaO2 / PCO2** | ~150 / normocapnic |
| **Lactate** | Falling; cleared by 12–24 h |
| **Limb** | Doppler signals, NIRS > 50% and symmetric |
| **LV** | Ejecting, not dilating, no smoke, lungs not filling |

## VV SELECTION

| Criterion | Threshold | Source |
|---|---|---|
| **Hypoxaemia** | PaO2/FiO2 < 80 for > 6 h, or < 50 for > 3 h, despite FiO2 ≥ 0.8, PEEP ≥ 10, Vt 6 mL/kg and proning where possible | EOLIA / ELSO 2021 |
| **Hypercapnia** | pH < 7.25 with PaCO2 ≥ 60 for > 6 h despite rate 35 and plateau ≤ 30–32 | EOLIA (≤ 32) / ELSO (≤ 30) |
| **Severity scores** | Murray lung injury score ≥ 3 (CESAR); RESP score for prognosis (not selection) | CESAR; Schmidt 2014 |
| **Other** | Bridge to lung transplant; air leak; status asthmaticus with injurious ventilation; airway surgery; DAH; severe primary graft dysfunction | ELSO |
| **Call early** | The durations are trial enrolment windows, not waiting times: a patient on 100% oxygen and 30 cm H2O getting worse is called now | ELSO |

## VA SELECTION, ECPR, HYPOTHERMIA

| Setting | Criteria and notes |
|---|---|
| **Refractory cardiogenic shock** | SBP < 90 (or vasopressors), CI < 2.2 with PCWP > 15, lactate > 2, SvO2 < 60%, oliguria, altered mentation despite optimal treatment for hours — SCAI stage C–E. Consider within 6 h of refractoriness, before the second organ fails. |
| **Aetiologies that recover** | Fulminant myocarditis, infarct (after revascularisation — ECLS-SHOCK says routine early VA does not help), post-cardiotomy, peripartum cardiomyopathy, intoxication (β-blocker, calcium-channel blocker, TCA), hypothermia, massive PE (2026 PE guideline: reasonable rescue), RV failure, refractory arrhythmia storm, primary graft failure. |
| **ECPR** | Witnessed arrest, bystander CPR (no-flow < 5 min), shockable rhythm or PEA, age < 70, ETCO2 > 10 during CPR, arrest-to-flow < 60 min, a presumed reversible cause, signs of life; cannulate after 10–20 min of failed resuscitation. ELSO registry survival 29%. |
| **Accidental hypothermia** | Arrest with core < 30 °C: ECLS rewarming if HOPE-predicted survival ≥ 10% (age, sex, temperature, potassium, CPR duration, asphyxia); normothermic exclusions do not apply. |
| **Contraindications** | Severe aortic regurgitation; aortic dissection; no recovery and no transplant/LVAD path; severe peripheral vascular disease (use the axillary or central route); unwitnessed or prolonged arrest; devastating brain injury; Child–Pugh B/C cirrhosis; a contraindication to anticoagulation is relative. |
| **SAVE score** | Diagnosis, age, weight, organ failures, pre-ECMO arrest, diastolic pressure, pulse pressure, bicarbonate, duration of ventilation, peak pressure: survival 75% (class I) → 18% (class V). |

## ANTICOAGULATION

| | Practice |
|---|---|
| **Drug** | Unfractionated heparin: 50–100 U/kg bolus at cannulation, then 10–20 U/kg/h titrated. Bivalirudin (renal clearance) or argatroban (hepatic) in HIT or heparin resistance. |
| **Targets (UFH)** | Anti-Xa 0.3–0.7 IU/mL (0.3–0.5 for VV at many centres; EOLIA used 0.2–0.3); aPTT 1.5–2.5 × baseline or ~60–80 s (reagent-dependent: do not transfer seconds between hospitals); ACT 180–220 at the bedside. RATE 2026: lower-dose UFH (aPTT 1.5–2×) non-inferior. |
| **Monitoring** | ACT q1–2 h early; anti-Xa/aPTT q6–12 h; platelets q6–12 h; fibrinogen, INR, CBC q12–24 h; plasma free Hb daily; D-dimer; antithrombin if heparin-resistant (> 50–80%); TEG/ROTEM when bleeding or clotting. |
| **Anti-Xa pitfalls** | Falsely low with plasma free Hb > 50 mg/dL, bilirubin > 6, triglycerides > 500; discordant aPTT (high) with anti-Xa (low) = factor deficiency, not more heparin. |
| **Thresholds** | Not bleeding: platelets ≥ 50 (some 30–50), fibrinogen > 1.0–1.5 g/L, INR < 3, Hb ≥ 7. Bleeding or procedure: platelets ≥ 100, fibrinogen > 1.5–2, INR < 1.5, Hb 7–9. |
| **Holding** | For bleeding: hold 4–6 h (up to 12 h or longer), keep the flow ≥ 3 L/min, inspect the circuit; VA needs it back sooner than VV. ICH on VV: hold > 2 days. Decannulation: hold ≥ 1 h. |
| **HIT** | Thrombocytopenia on ECMO is usually not HIT (consumption, sepsis, drugs): 4T score and PF4/serotonin release; if HIT, stop all heparin including the coating if possible and use a DTI. |

## DAILY LABS

| Test | Frequency | Meaning |
|---|---|---|
| **Pre- and post-membrane gas** | Daily and after changes | Transfer, membrane function, recirculation (pre-membrane saturation) |
| **Plasma free Hb** | Daily | < 10 normal; 50–100 moderate (find the cause); > 100 severe (act: speed, pump head, clot) |
| **LDH, haptoglobin, bilirubin, potassium** | Daily | Haemolysis corroboration; renal risk |
| **Platelets, fibrinogen, D-dimer** | q12–24 h | Consumption in the circuit; rising D-dimer predicts membrane failure |
| **ΔP at a stated flow** | Every shift | Clot in the membrane |
| **Lactate, SvO2** | q6–12 h | Delivery versus demand |
| **Chest X-ray** | Daily early | Cannula position, lung recovery, pneumothorax, oedema |
| **Echo** | Daily on VA; as needed on VV | LV ejection and distension, RV, tamponade, cannula positions, recirculation jet |
| **Neurological check** | Every shift; pupillometry; NIRS on VA | Stroke and haemorrhage are the deaths on ECMO; sedation holidays when safe |
| **Limb check (VA)** | Hourly | DPC flow, Doppler, NIRS, compartments |

## THE EMERGENCIES

| Event | Recognise | Do |
|---|---|---|
| **Air in the circuit** | Bubbles in the line, bubble-detector alarm, sudden flow loss, pump cavitating; the source is a pre-pump port, a cracked connector, a cannula pulled back | Clamp return → clamp drainage → pump to zero; head down (left lateral if possible); FiO2 1.0 and emergency ventilator settings; pressors on VA; perfusionist de-airs (two-syringe) or changes the circuit; restart drainage-unclamp, speed, return-unclamp |
| **Pump failure / power loss** | Console dark, flow 0; on VA the flow reverses (arterio-venous fistula) | Clamp (VA), hand crank, backup console, restore rpm; support the patient; charge/replace |
| **Accidental decannulation** | Bleeding from the site, air, loss of flow | Clamp, pump off, direct pressure (the vein bleeds less than you fear, the artery more), rescue ventilation and pressors, recannulate if still needed |
| **Suck-down / drainage insufficiency** | Chatter, flow swings, drainage pressure < −100, alarms | Speed down 100 rpm at a time until stable; then volume status, kink, position (X-ray/echo), cough, tamponade, pneumothorax, abdomen |
| **Oxygenator failure** | Post-membrane PO2 < 200 on FdO2 1.0, PCO2 clearance lost, ΔP rising at constant flow, plasma leak, consumption/haemolysis | Check the gas line first; elective exchange prepared (prime, rescue settings, team) before it fails |
| **Pump-head thrombosis** | Noise, heat, sudden haemolysis, flow falling at the same rpm, ΔP across the head | Change the pump head (or circuit) |
| **Massive haemolysis** | Free Hb > 100, dark urine, hyperkalaemia, AKI | Reduce suction and speed; find the clot or kink; change the component; protect the kidney |
| **Tamponade on VA** | Loss of pulsatility, rising CVP, falling flow with chatter, echo | Pericardiocentesis or the surgeon |
| **Intracranial haemorrhage** | New deficit, seizure, pupil change, failure to wake; CT | Hold anticoagulation (> 2 days on VV), platelets > 100, neurosurgery; the circuit can run unanticoagulated at full flow |
| **Limb ischaemia** | Cold, pale, no Doppler, NIRS < 50% or > 15–20 points asymmetric; later tense compartments, CK, potassium | Distal perfusion catheter; vascular surgery; fasciotomy |
| **Pneumothorax on VV** | Desaturation, chatter (pleural pressure), flow loss | Ultrasound/X-ray; drain carefully (anticoagulated); small-bore |
| **Sweep disconnection** | Sudden desaturation and hypercapnia with a normal ΔP; the exhaust silent | Reconnect; it is the first check |

## BLEEDING

- Where: cannula sites (most), surgery, airway and lung, GI, retroperitoneum, intracranial. Hb trend, platelets, fibrinogen, INR, anti-Xa/aPTT; TEG/ROTEM.
- Reduce or hold anticoagulation (4–6 h, longer if needed); correct: platelets ≥ 100, fibrinogen > 1.5–2 g/L, INR < 1.5, calcium; keep the flow ≥ 3 L/min; antifibrinolytics for life-threatening bleeding; DDAVP for acquired von Willebrand syndrome (shear destroys the large multimers on every circuit); source control (surgery, endoscopy, embolisation).
- Tracheostomy, chest drains and bronchoscopy are elective bleeding: plan the anticoagulation around them.
- Do not stop the circuit for bleeding; do stop the heparin.

## VV: THE SWEEP-OFF TRIAL

| Step | What |
|---|---|
| **Readiness** | The lung is recovering (X-ray, compliance, PaO2/FiO2 at a lower FdO2), haemodynamics stable, secretions manageable, drive acceptable. ELSO: FiO2 ≤ 0.6, PEEP ≤ 10, PaO2 ≥ 70; Vt ≤ 6 mL/kg, plateau ≤ 28, RR ≤ 28 on the ventilator. |
| **Wean the gas** | FdO2 from 1.0 toward 0.21 in steps of ~0.2 keeping SpO2 > 92% / PaO2 ≥ 70; then sweep by 0.5–1 L/min steps toward 1 L/min with a gas at each step; tolerating sweep < 2 L/min is the gate. Keep blood flow > 2–3 L/min (> 1 L/min per cannula) throughout. |
| **The trial** | Sweep to zero, gas line capped; the blood still circulates. Ventilator at the acceptable settings (not escalated to pass). ABG at 1–2 h and clinically for 2–3 h or longer (a 30-min gas is a screen; ELSO watches hours). Pass: PaO2 ≥ 70 on FiO2 ≤ 0.6, pH ≥ 7.3, PaCO2 acceptable, RR ≤ 28, plateau ≤ 28, no distress, haemodynamics unchanged. |
| **Fail** | Sweep back on; find why (fluid, secretions, effort, infection); retry in a day. |
| **Decannulate** | Heparin off ≥ 1 h; the venous cannulas come out with manual compression or a purse-string, head down for the jugular (air), positive-pressure breath-hold or brief paralysis; compression 20–30 min; ultrasound the veins after 24 h for DVT (half have one). |
| **After** | The lung is still convalescent: lung-protective ventilation continues; most patients are extubated after decannulation, not before. |

## VA: THE FLOW-DOWN TRIAL

| Step | What |
|---|---|
| **Readiness** | Cause treated or recovering; pressors off or low (norepinephrine ≤ 0.06–0.1 µg/kg/min, no high-dose inotrope); lactate normal; pulsatile trace with pulse pressure > 20–30; echo at full flow improving; lungs adequate on the ventilator (the native lung must oxygenate once the circuit goes); no tamponade; rhythm stable; LVEF ≥ 20–25% and rising. Usually day 3–7. |
| **The trial** | Heparin bolus (or ACT > 200). Reduce flow in 0.5 L/min steps every 5–10 min (sweep kept on; FdO2 up if needed) to 1–1.5 L/min (never lower for long: thrombosis); echo and haemodynamics at each step; hold 30–60 min at minimal flow. Some centres add a 3–5-min clamp with the bridge open, or a pump-controlled retrograde trial (flow reversed 1 L/min through the circuit with the sweep off, which mimics decannulation). |
| **Pass (Aissaoui criteria at minimal flow)** | MAP ≥ 60–65 without new vasopressors; LVOT VTI ≥ 10 cm (≥ 12 per ELSO); LVEF > 20–25%; lateral mitral annular TDSa ≥ 6 cm/s; no LV dilation or rising filling pressures (CVP ≤ 10); SvO2 > 65%; no arrhythmia, no desaturation; lactate stable. |
| **Fail** | Back to the last tolerated flow; address the cause (volume, ischaemia, rhythm, valve); if no recovery by 5–7 days, the next device (temporary LVAD, durable LVAD, transplant listing) or the limit of the trial. |
| **Decannulate** | Promptly after a pass (the circuit at 1 L/min is a clot waiting). Arterial: surgical repair, or percutaneous closure devices; DPC out; venous as for VV. Watch the leg and the groin (pseudoaneurysm, AV fistula, retroperitoneal bleed). |
| **Heart recovered, lungs not** | Convert VA → VV (swap the arterial return for a jugular venous return) rather than forcing full decannulation. |

## THE TWO MISTAKES

- Sweep off on VA: with the gas off the circuit returns venous blood straight into the artery — a right-to-left shunt of the whole flow. VA weaning never turns the sweep off while blood goes to an artery (the retrograde trial is the deliberate exception, at 1 L/min, for minutes).
- Flow down on VV: the membrane still works at 1 L/min and the circuit clots; it tests nothing about the lung. VV weaning is the sweep.

## THE ROUNDS CHECKLIST

1. Purpose: what is the disease, what is the bridge, what changed today?
2. Support: mode and sites; flow at what rpm; sweep and FdO2; drainage and membrane pressures and their trend; ventilator settings and the native contribution.
3. Delivery: arterial (right radial on VA) and post-membrane gases, Hb, lactate, SvO2, urine, perfusion, echo.
4. Safety: cannula depths and dressings; ΔP and transfer; bleeding, clot, haemolysis; neurology; limbs; anticoagulant and assay.
5. Recovery: fluid balance, sedation and effort, nutrition, infection, mobility, organ trajectory.
6. Exit: readiness for a weaning trial (VV sweep-off; VA flow-down), the next decision point, the family.

Sources: Tonna JE et al, ELSO adult VV guideline, ASAIO J 2021 · Lorusso R et al, ELSO interim VA guideline, ASAIO J 2021 · McMichael ABV et al, ELSO anticoagulation guidelines, ASAIO J 2022 · Richardson AS et al, ELSO ECPR consensus, ASAIO J 2021 · ELSO neurological monitoring consensus, ASAIO J 2024 · ELSO hypothermia narrative guideline 2025 · Combes (EOLIA 2018), Peek (CESAR 2009), Thiele (ECLS-SHOCK 2023), Schmidt (PRONECMO 2023), McNamee (REST 2021), Yannopoulos (ARREST 2020), Suverein (INCEPTION 2023), van Minnen (RATE 2026), Aissaoui (ICM 2011) · Farkas, IBCC: ECMO (2023).